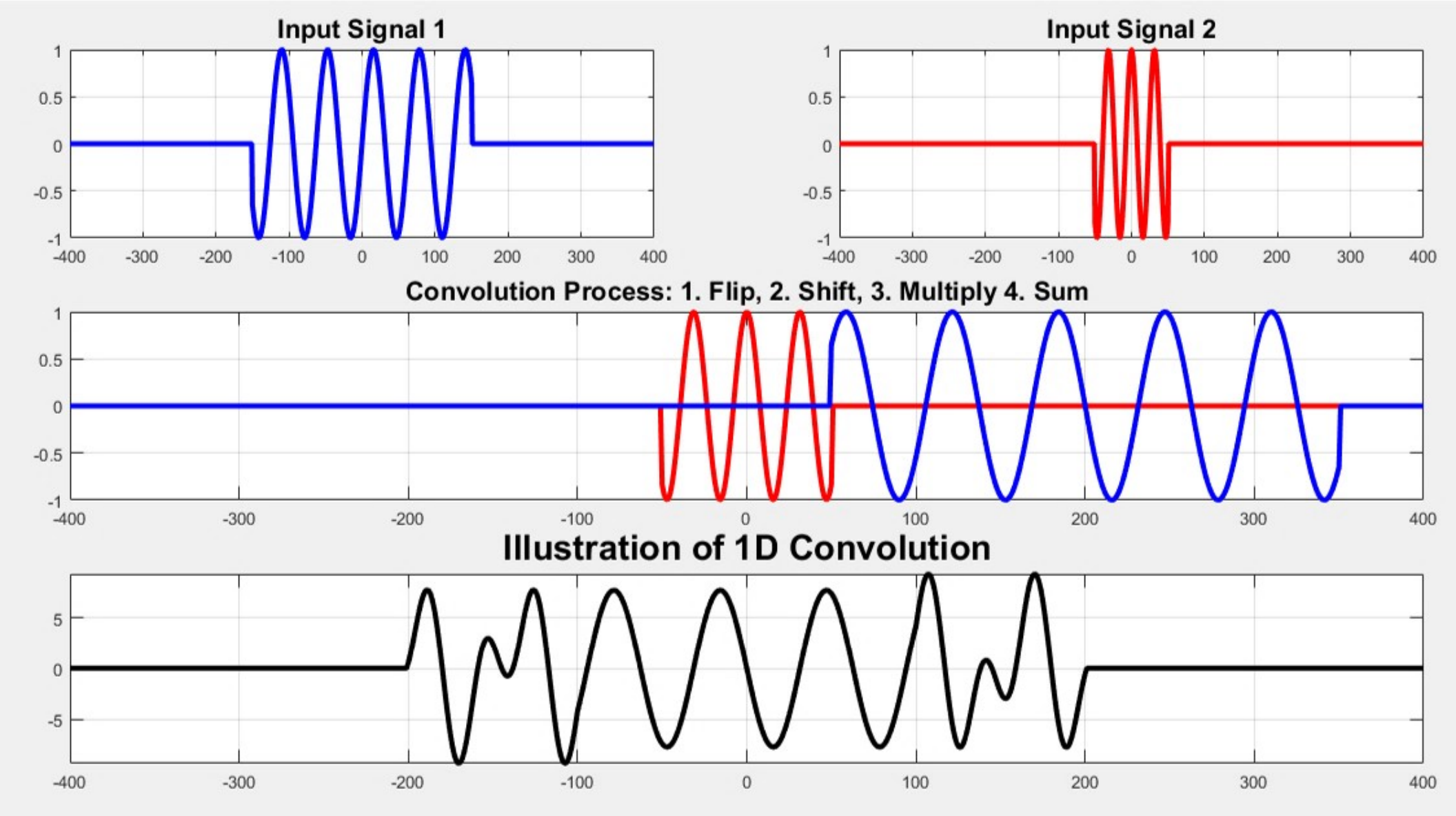
Editor - C:\Users\Hamza Khalid\Documents\Matlab Script Files\DSP Lab Tasks\DSP Lab Project\mainfile.m

Command Window

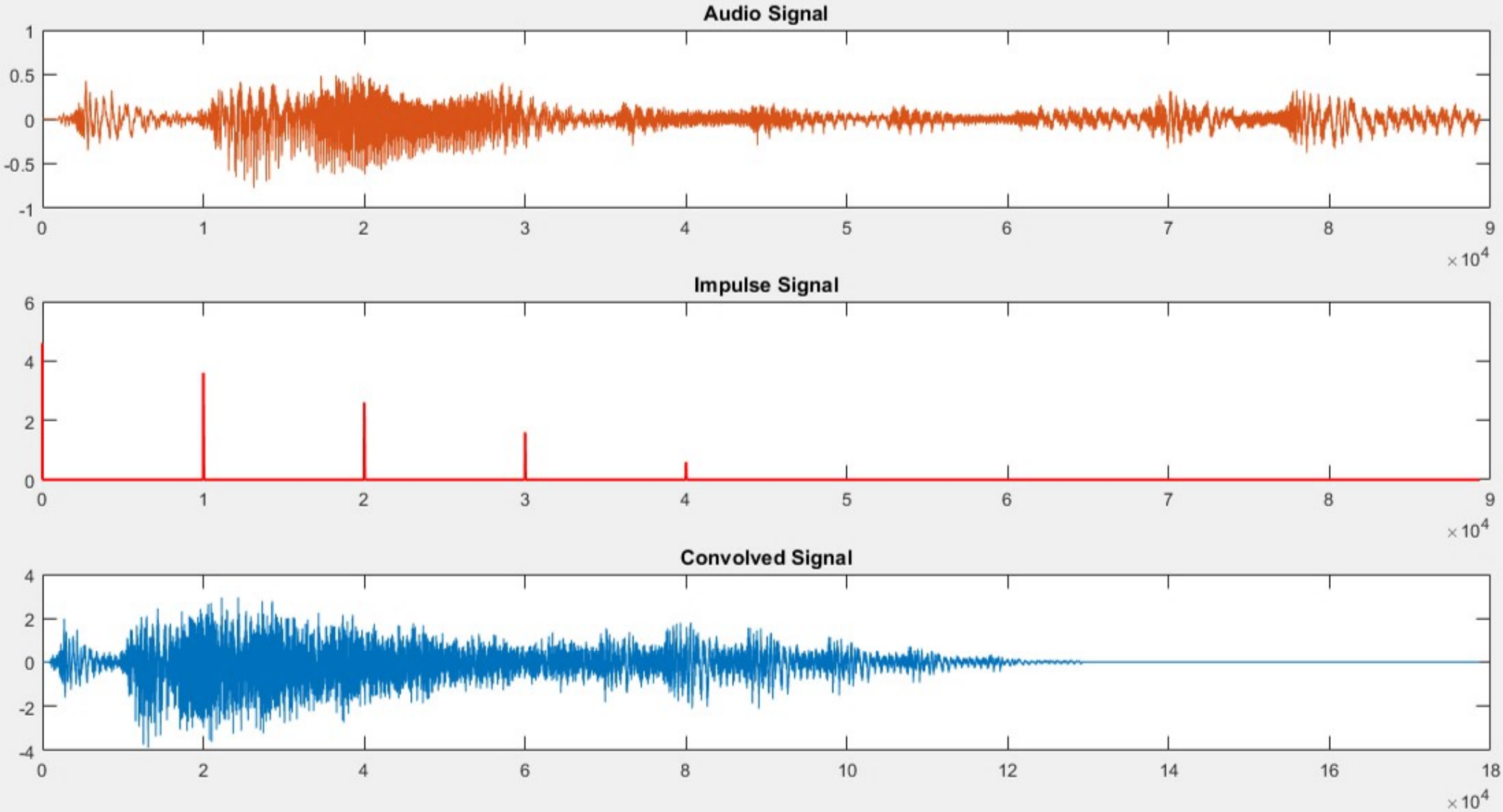
```
DSP PROJECT
Convolution & Quantization
*****************************************
Menu:
1) Convolution
2) Quantization
*****************************************
Enter Your Choice: 1
*****************************************
Convolution SubMenu:
1) Signal Convolution
2) Audio Convolution (Echo)
3) Image Convolution (Blur)
*****************************************
Enter Your Choice: 1
```


Input Signal 1
Input Signal 2
Convolution Process: 1. Flip, 2. Shift, 3. Multiply 4. Sum
Illustration of 1D Convolution

Editor - C:\Users\Hamza Khalid\Documents\Matlab Script Files\DSP Lab Tasks\DSP Lab Project\mainfile.m

Command Window

```
DSP PROJECT
Convolution & Quantization
*****************************************
Menu:
1) Convolution
2) Quantization
*****************************************
Enter Your Choice: 1
*****************************************
Convolution SubMenu:
1) Signal Convolution
2) Audio Convolution (Echo)
3) Image Convolution (Blur)
*****************************************
Enter Your Choice: 2
>>
```


Audio Signal
1
0.5
0
-0.5
-1
0
1
2
3
4
5
6
7
8
9
×10^4
Impulse Signal
6
4
2
0
0
1
2
3
4
5
6
7
8
9
×10^4
Convolved Signal
4
2
0
-2
-4
0
2
4
6
8
10
12
14
16
18
×10^4

Editor - C:\Users\Hamza Khalid\Documents\Matlab Script Files\DSP Lab Tasks\DSP Lab Project\mainfile.m

Command Window

```
DSP PROJECT
Convolution & Quantization
*****************************************
Menu:
1) Convolution
2) Quantization
*****************************************
Enter Your Choice: 1
*****************************************
Convolution SubMenu:
1) Signal Convolution
2) Audio Convolution (Echo)
3) Image Convolution (Blur)
*****************************************
Enter Your Choice: 3
```

Original Image

Convolution with Gaussian Filter

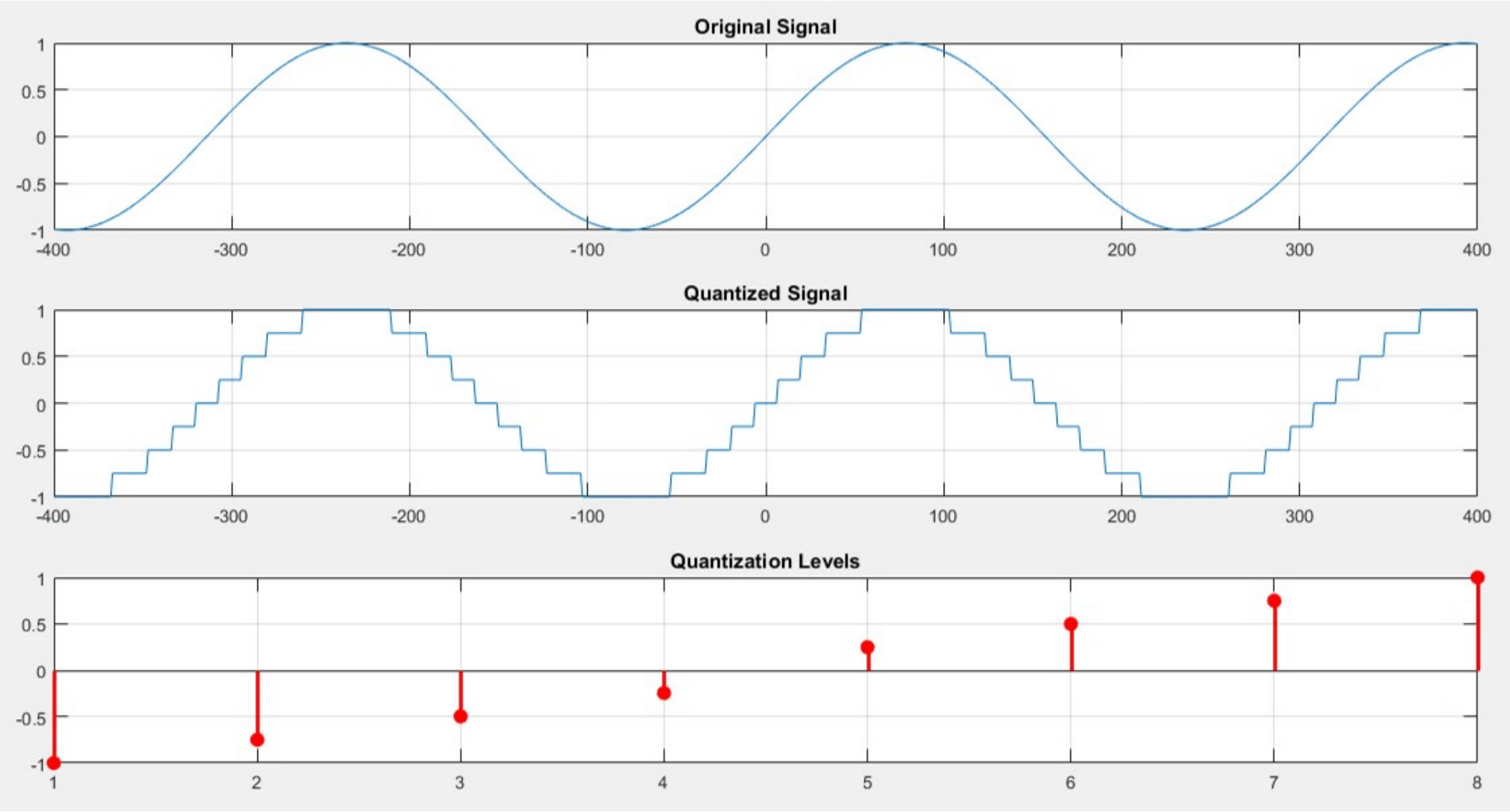
Convolution with Motion Filter

Self Convolution

Editor - C:\Users\Hamza Khalid\Documents\Matlab Script Files\DSP Lab Tasks\DSP Lab Project\mainfile.m

Command Window

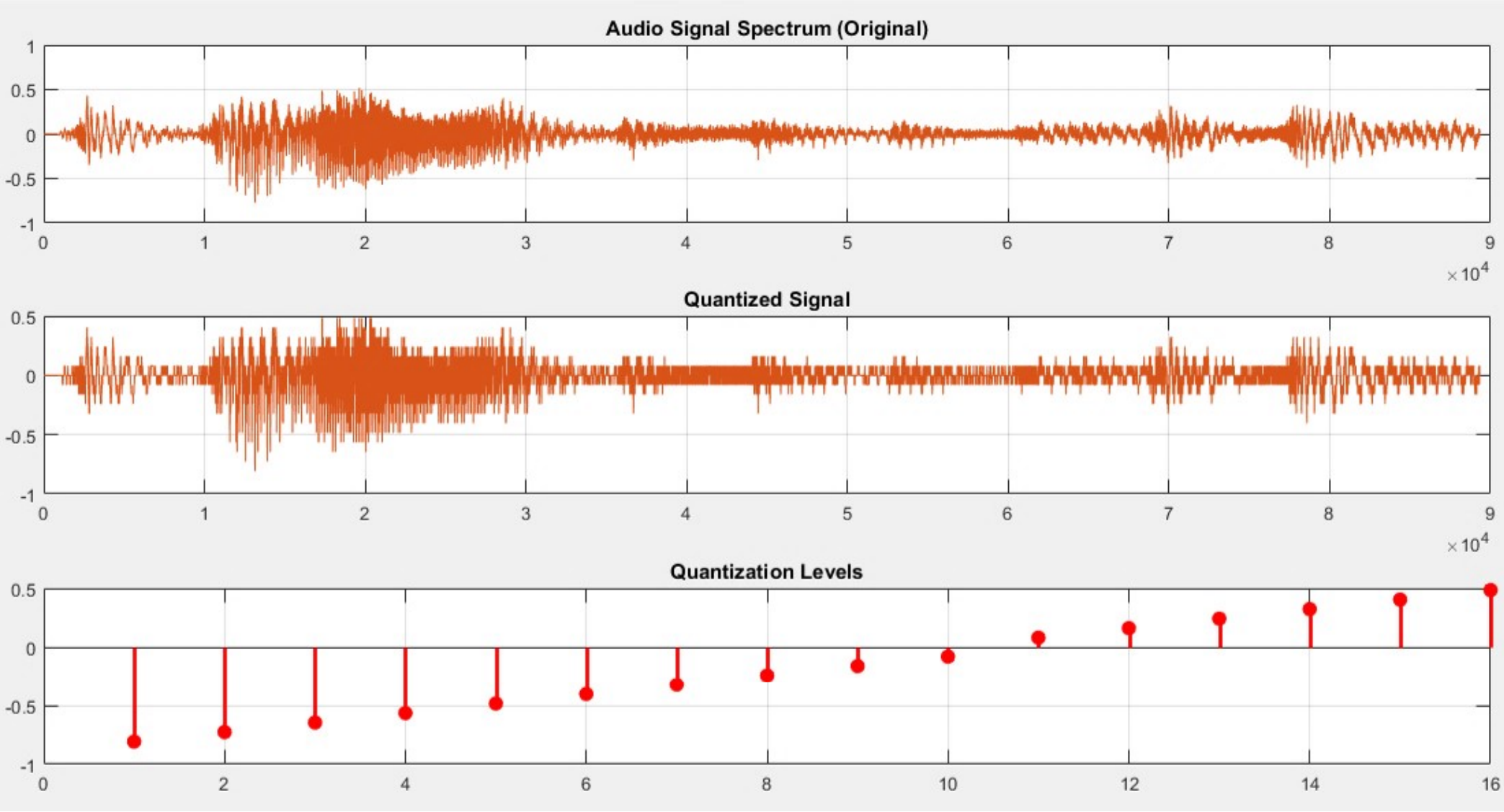
```
DSP PROJECT
Convolution & Quantization
*****************************************
Menu:
1) Convolution
2) Quantization
*****************************************
Enter Your Choice: 2
*****************************************
Quantization SubMenu:
1) Signal Quantization
2) Audio Quantization
3) Image Quantization
*****************************************
Enter Your Choice: 1
>>
```


Original Signal
1
0.5
0
-0.5
-1
-400
-300
-200
-100
0
100
200
300
400
Quantized Signal
1
0.5
0
-0.5
-1
-400
-300
-200
-100
0
100
200
300
400
Quantization Levels
1
0.5
0
-0.5
-1
1
2
3
4
5
6
7
8

Editor - C:\Users\Hamza Khalid\Documents\Matlab Script Files\DSP Lab Tasks\DSP Lab Project\mainfile.m

Command Window

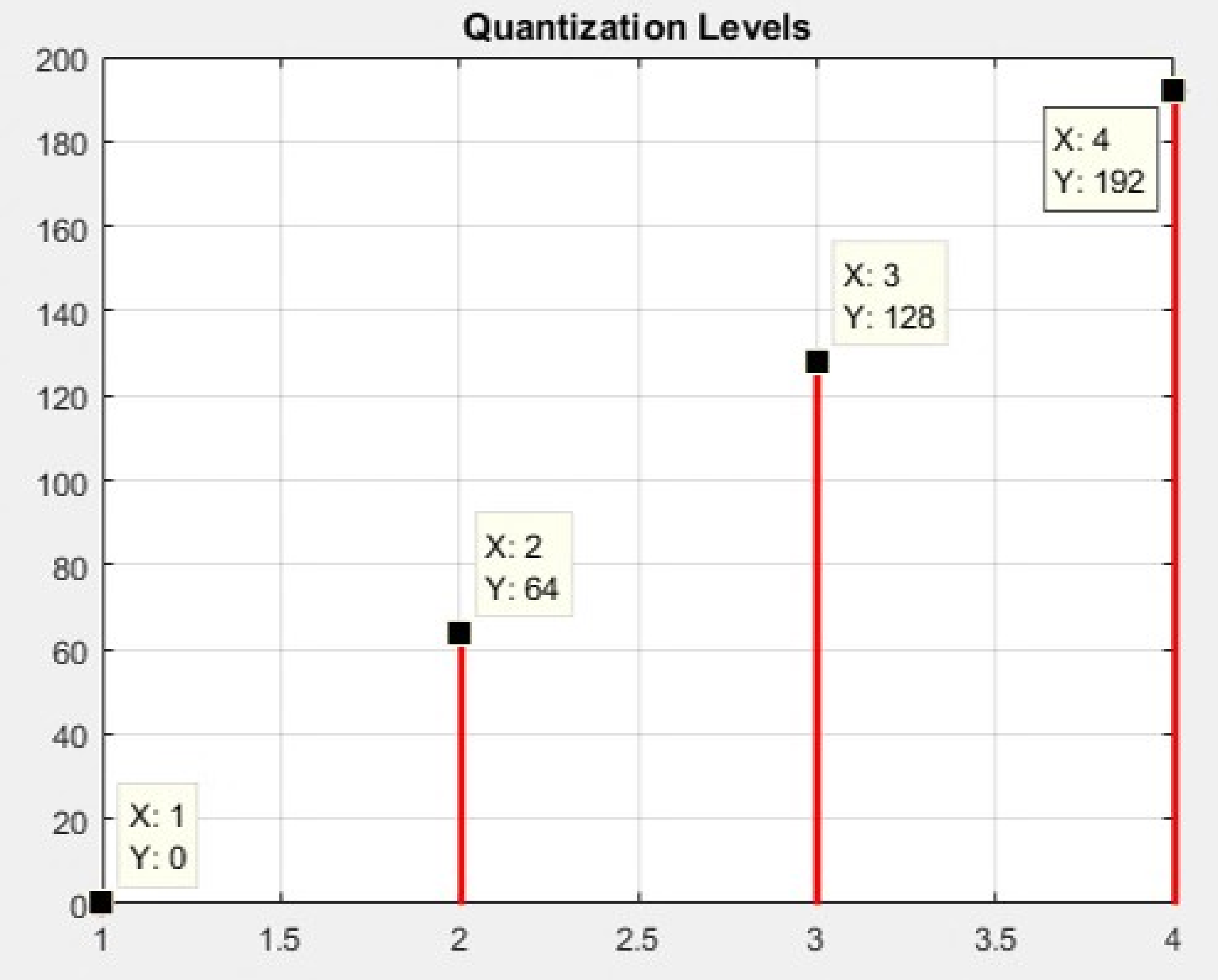
```
DSP PROJECT
Convolution & Quantization
*****************************************
Menu:
1) Convolution
2) Quantization
*****************************************
Enter Your Choice: 2
*****************************************
Quantization SubMenu:
1) Signal Quantization
2) Audio Quantization
3) Image Quantization
*****************************************
Enter Your Choice: 2
>>
```


Audio Signal Spectrum (Original)
1
0.5
0
-0.5
-1
0
1
2
3
4
5
6
7
8
9
×10^4
Quantized Signal
0.5
0
-0.5
-1
0
1
2
3
4
5
6
7
8
9
×10^4
Quantization Levels
0.5
0
-0.5
-1
0
2
4
6
8
10
12
14
16

Editor - C:\Users\Hamza Khalid\Documents\Matlab Script Files\DSP Lab Tasks\DSP Lab Project\mainfile.m

Command Window

```
DSP PROJECT
Convolution & Quantization
*****************************************
Menu:
1) Convolution
2) Quantization
*****************************************
Enter Your Choice: 2
*****************************************
Quantization SubMenu:
1) Signal Quantization
2) Audio Quantization
3) Image Quantization
*****************************************
Enter Your Choice: 3
>>
```


Quantization Levels
200
180
160
140
120
100
80
60
40
20
0
1
1.5
2
2.5
3
3.5
4
X: 1
Y: 0
X: 2
Y: 64
X: 3
Y: 128
X: 4
Y: 192

Original Image

Quantized Image